

```

In [2]: import pandas as pd
import numpy as np

In [3]: bookings = pd.read_csv("bookings.csv")
rooms = pd.read_csv("rooms.csv")
guests = pd.read_csv("guests.csv")

In [4]: bookings.duplicated().sum()

Out[4]: np.int64(8)

In [5]: bookings = bookings.drop_duplicates()

In [6]: bookings.duplicated().sum()

Out[6]: np.int64(0)

In [7]: bookings.head(0)

Out[7]: BookingID  CheckinDate  GuestID  RoomType  Nights  GuestsCount  TotalAmount  Source  Rating

In [8]: guests.head(0)

Out[8]: GuestID  GuestName  City  Age  Phone

In [9]: rooms.head(0)

Out[9]: RoomType  Base_Price  Max_Occupancy

In [10]: bookings.columns = bookings.columns.str.title().str.strip().str.lower()
guests.columns = guests.columns.str.title().str.strip().str.lower()
rooms.columns = rooms.columns.str.title().str.strip().str.lower()

In [11]: bookings.isnull().sum()

Out[11]: bookingid      0
checkindate      0
guestid          0
roomtype         0
nights           69
guestscount      55
totalamount      83
source           0
rating           39
dtype: int64

In [12]: # removing null values
bookings['nights'] = bookings['nights'].fillna(bookings['nights'].median())
bookings['guestscount'] = bookings.groupby('roomtype')['guestscount'].transform(lambda x: x.fillna(x.mode()[0])

In [13]: bookings.isnull().sum()

Out[13]: bookingid      0
checkindate      0
guestid          0
roomtype         0
nights           0
guestscount      55
totalamount      83
source           0
rating           39
dtype: int64

In [14]: # removing guestcount null values using by there room type
bookings['guestscount'] = bookings.groupby('roomtype')['guestscount'].transform(lambda x: x.fillna(x.mode()[0])

In [15]: rooms['roomtype'] = rooms['roomtype'].str.title().str.strip().str.lower()

In [16]: bookings['roomtype'] = bookings['roomtype'].str.title().str.strip().str.lower()
bookings['guestscount'] = bookings['guestscount'].str.title().str.strip().str.lower()
bookings['source'] = bookings['source'].str.title().str.strip().str.lower()

In [17]: guests.head()

```

Out[17]:

	guestid	guestname	city	age	phone
0	2001	Sameer Kumar	Noida	23.0	9861233189
1	2002	Ankit Sharma	Delhi	44.0	9440132810
2	2003	Simran Verma	Noida	22.0	9951505986
3	2004	Divya Sharma	delhi	38.0	9231757328
4	2005	Yash Singh	Noida	42.0	9988872530

In [18]:

```
guests['guestname'] = guests['guestname'].str.title().str.strip().str.lower()  
guests['city'] = guests['city'].str.title().str.strip().str.lower()
```

In [19]:

```
bookings.isnull().sum()
```

Out[19]:

```
bookingid      0  
checkindate    0  
guestid        0  
roomtype       0  
nights         0  
guestscount    0  
totalamount    83  
source         0  
rating        39  
dtype: int64
```

In [20]:

```
bookings.head(1)
```

Out[20]:

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating
0	9000	26-10-2025	2025	suite	2.0	suite	12000.0	direct	5.0

In [21]:

```
rooms.head(1)
```

Out[21]:

	roomtype	base_price	max_occupancy
0	deluxe	3500	3

In [22]:

```
bookings = pd.merge(bookings,rooms[['roomtype','base_price']], on = 'roomtype' , how = 'left')  
bookings['totalamount'] = bookings['totalamount'].fillna(bookings['nights'] * bookings['base_price'])
```

In [23]:

```
bookings.isnull().sum()
```

Out[23]:

```
bookingid      0  
checkindate    0  
guestid        0  
roomtype       0  
nights         0  
guestscount    0  
totalamount    0  
source         0  
rating        39  
base_price     0  
dtype: int64
```

In [24]:

```
bookings['rating'] = bookings.groupby('roomtype')['rating'].transform(lambda x: x.fillna(x.mean()))
```

In [25]:

```
bookings.isnull().sum()
```

Out[25]:

```
bookingid      0  
checkindate    0  
guestid        0  
roomtype       0  
nights         0  
guestscount    0  
totalamount    0  
source         0  
rating         0  
base_price     0  
dtype: int64
```

In [26]:

```
guests.isnull().sum()
```

Out[26]:

guestid	0
guestname	0
city	0
age	2
phone	0
dtype:	int64

In [27]:

```
guests['age'] = guests['age'].fillna(guests['age'].median())
```

In [28]:

```
# calculate total revenue using roomtype and total amount
bookings.groupby('roomtype')['totalamount'].sum().sort_values(ascending = False)
```

Out[28]:

roomtype	
suite	1308000.0
cottage	693000.0
deluxe	560000.0
standard	534000.0
dormitory	127800.0
Name: totalamount, dtype:	float64

In [29]:

```
bookings.groupby('source').agg(
    total_bookings = ('bookingid','count'),
    avg_amount = ('totalamount','mean'),
    avg_rating = ('rating','mean')
).sort_values('total_bookings', ascending = False)
```

Out[29]:

	total_bookings	avg_amount	avg_rating
source			
direct	140	9170.714286	3.834337
makemytrip	109	8489.908257	3.912946
goibibo	53	8349.056604	3.948344
booking.com	50	11420.000000	3.823087

In [30]:

```
bookings.head(0)
```

Out[30]:

bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
-----------	-------------	---------	----------	--------	-------------	-------------	--------	--------	------------

In [31]:

```
guests.head(0)
```

Out[31]:

guestid	guestname	city	age	phone
---------	-----------	------	-----	-------

In [32]:

```
# merge to table
full_data = pd.merge(bookings,guests ,on = 'guestid' , how = 'left')
```

In [33]:

```
full_data.head()
```

Out[33]:

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
0	9000	26-10-2025	2025	suite	2.0	suite	12000.0	direct	5.0	6000
1	9001	01-01-2025	2035	suite	1.0	suite	6000.0	booking.com	5.0	6000
2	9002	27-11-2025	2031	dormitory	2.0	dormitory	1800.0	goibibo	4.0	900
3	9003	15-01-2025	2046	suite	4.0	suite	24000.0	direct	4.0	6000
4	9004	13-09-2025	2006	suite	4.0	suite	24000.0	makemytrip	4.0	6000

In [34]:

```
# city wise total revenue
full_data.groupby('city')['totalamount'].sum().sort_values(ascending = False)
```

Out[34]:

city	
delhi	772400.0
noida	700300.0
chandigarh	511100.0
lucknow	487900.0
dehradun	424800.0
meerut	326300.0
Name: totalamount, dtype:	float64

In [35]:

```
# making a pivot table
pd.pivot_table(full_data, values = 'totalamount', index = 'roomtype', columns = 'source',aggfunc = 'sum', fill_
```

Out[35]:

	source	booking.com	direct	goibibo	makemytrip
roomtype					
cottage		31500.0	297000.0	139500.0	225000.0
deluxe		91000.0	220500.0	98000.0	150500.0
dormitory		22500.0	50400.0	9000.0	45900.0
standard		180000.0	218000.0	46000.0	90000.0
suite		246000.0	498000.0	150000.0	414000.0

In [36]: `# top 5 guests`
`full_data.groupby('guestname')['totalamount'].sum().sort_values(ascending = False).head(5)`

Out[36]:

```
guestname
ravi verma      243200.0
karan chauhan    159400.0
karan verma     122000.0
divya sharma    120800.0
pooja singh     115000.0
Name: totalamount, dtype: float64
```

In [37]: `# how many times guest stay in the hotel`
`full_data.groupby('guestname')['bookingid'].count().sort_values(ascending = False).head(5)`

Out[37]:

```
guestname
ravi verma      18
rohan mishra    17
simran verma    13
ravi kumar      11
sameer mishra   11
Name: bookingid, dtype: int64
```

In [38]: `bookings.head(2)`

Out[38]:

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
0	9000	26-10-2025	2025	suite	2.0	suite	12000.0	direct	5.0	6000
1	9001	01-01-2025	2035	suite	1.0	suite	6000.0	booking.com	5.0	6000

In [61]: `# creating a month and weekday column in the table`
`bookings['checkindate'] = pd.to_datetime(bookings['checkindate'])`
`bookings['month'] = bookings['checkindate'].dt.month`
`bookings['weekday'] = bookings['checkindate'].dt.day_name()`

In [62]: `bookings.head()`

Out[62]:

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
0	9000	2025-10-26	2025	suite	2.0	suite	12000.0	direct	5.0	6000
1	9001	2025-01-01	2035	suite	1.0	suite	6000.0	booking.com	5.0	6000
2	9002	2025-11-27	2031	dormitory	2.0	dormitory	1800.0	goibibo	4.0	900
3	9003	2025-01-15	2046	suite	4.0	suite	24000.0	direct	4.0	6000
4	9004	2025-09-13	2006	suite	4.0	suite	24000.0	makemytrip	4.0	6000

In [63]: `# calculate the which weekday has the most booking`
`bookings['weekday'].value_counts()`

Out[63]:

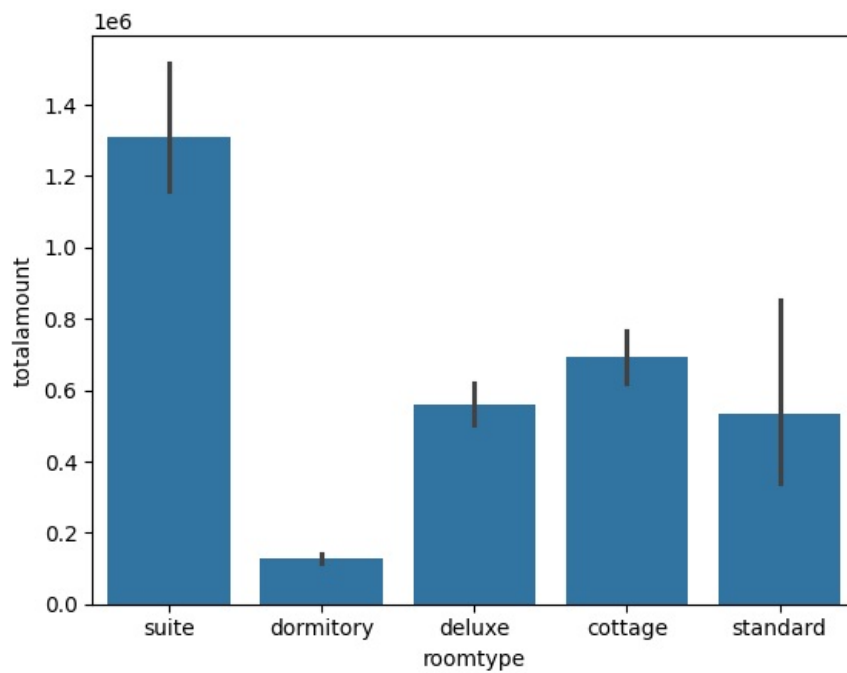
```
weekday
Saturday    58
Wednesday   57
Tuesday     53
Monday      53
Thursday    46
Sunday      44
Friday      41
Name: count, dtype: int64
```

In [64]: `# month wise total revenue`
`bookings.groupby('month')['totalamount'].sum()`

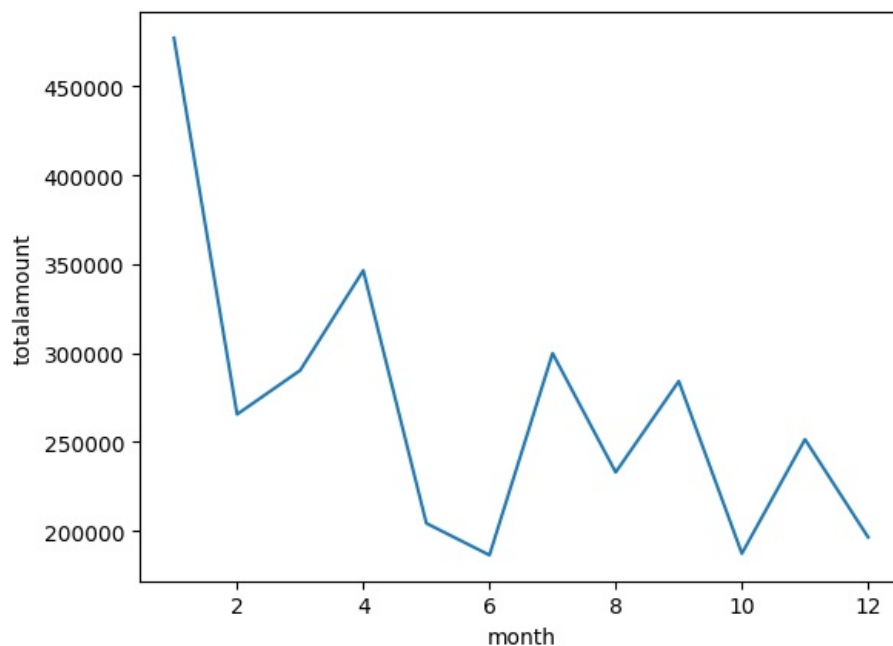
```
Out[64]: month
1      477300.0
2      265600.0
3      290300.0
4      346500.0
5      204300.0
6      186400.0
7      299900.0
8      233000.0
9      284200.0
10     187300.0
11     251500.0
12     196500.0
Name: totalamount, dtype: float64
```

```
In [65]: import seaborn as sns
import matplotlib.pyplot as plt
```

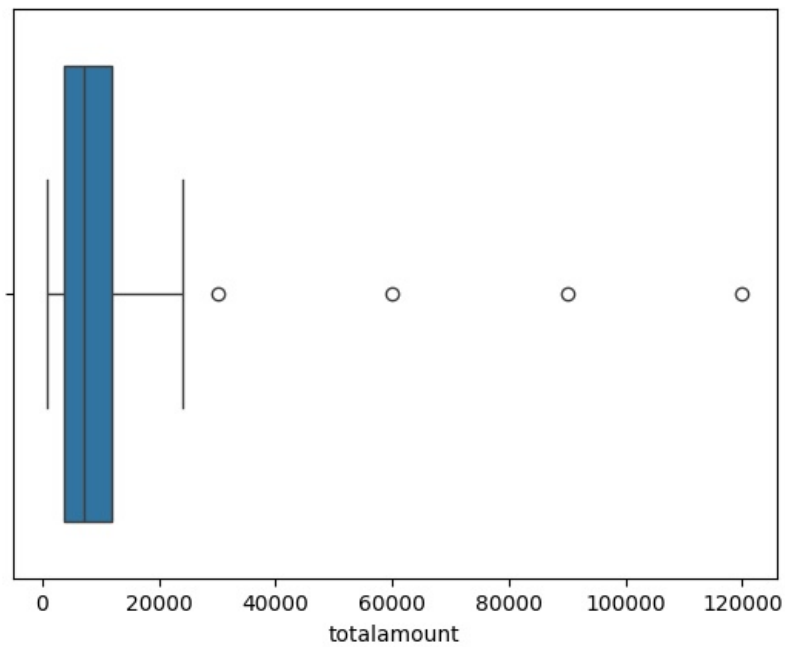
```
In [66]: # making a bar plot of room type wise total revenue
sns.barplot(x='roomtype', y='totalamount', data=bookings, estimator=sum)
plt.show()
```



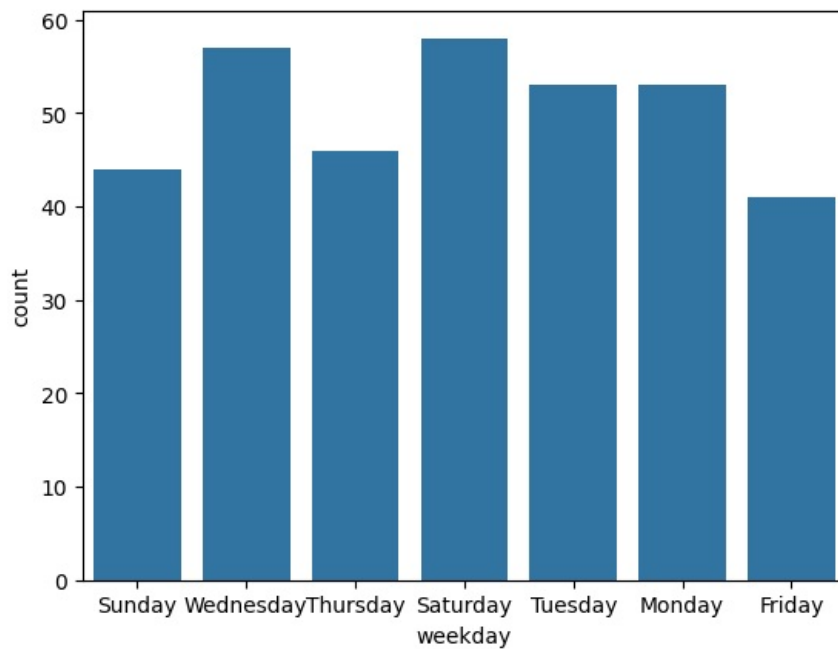
```
In [67]: monthly = bookings.groupby('month')['totalamount'].sum().reset_index()
sns.lineplot(x='month', y='totalamount', data=monthly)
plt.show()
```



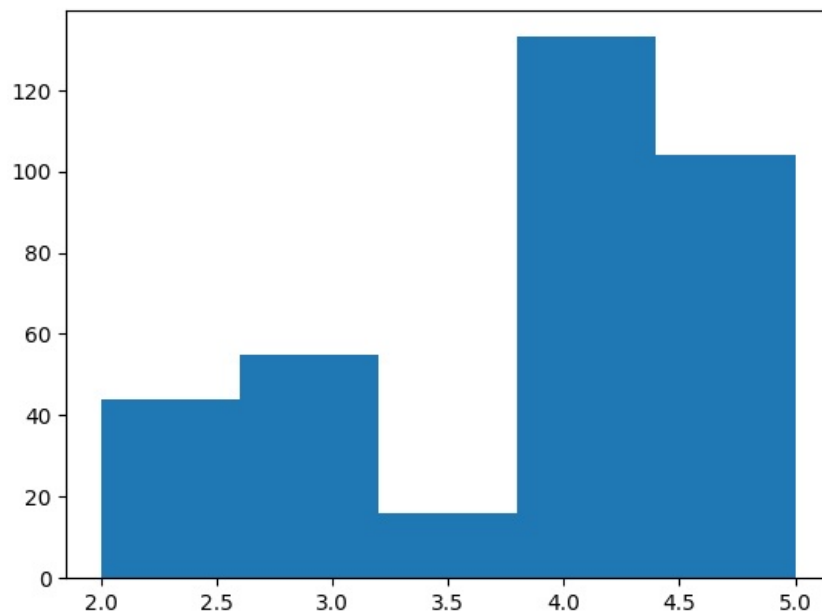
```
In [68]: # box plot of total amount
sns.boxplot(x=bookings['totalamount'])
plt.show()
```



```
In [69]: # week wise booking count
sns.countplot(x = 'weekday' , data = bookings)
plt.show()
```



```
In [70]: # histogram chart of rating
plt.hist(bookings['rating'], bins = 5)
plt.show()
```



```
In [71]: # calculate the total revenue using iqr
iq1 = bookings['totalamount'].quantile(0.25)
iq3 = bookings['totalamount'].quantile(0.75)
iqr = iq3 - iq1
outliers = bookings[(bookings['totalamount'] < iq1 - 1.5 * iqr) | (bookings['totalamount'] > iq3 + 1.5 * iqr)]
outliers
```

```
Out[71]:
```

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
15	9015	2025-02-25	2006	suite	1.0	suite	90000.0	booking.com	4.0	600
37	9037	2025-01-03	2057	standard	4.0	standard	120000.0	booking.com	3.0	200
66	9066	2025-07-24	2005	standard	1.0	standard	30000.0	direct	5.0	200
224	9224	2025-07-09	2047	standard	2.0	standard	60000.0	direct	5.0	200

```
In [72]: # create age group column
bins = [18,30,45,65]
labels = ['18-30','31-45','46-65']
guests['age_group'] = pd.cut(guests['age'], bins = bins, labels = labels)
```

```
In [73]: bookings.head(0)
```

```
Out[73]:
```

bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price	month
-----------	-------------	---------	----------	--------	-------------	-------------	--------	--------	------------	-------

```
In [74]: guests.head(0)
```

```
Out[74]:
```

guestid	guestname	city	age	phone	age_group
---------	-----------	------	-----	-------	-----------

```
In [75]: full_data2 = pd.merge(bookings,guests , on = 'guestid' , how = 'left')
```

```
In [76]: full_data2.groupby('age_group')['totalamount'].sum()
```

```
Out[76]: age_group
18-30      991900.0
31-45      862800.0
46-65     1368100.0
Name: totalamount, dtype: float64
```

```
In [77]: full_data2.head(1)
```

```
Out[77]:
```

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price	month
0	9000	2025-10-26	2025	suite	2.0	suite	12000.0	direct	5.0	6000	

```
In [78]: # calculate total revenue and find the rank using roomtype
revenue = full_data2.groupby('roomtype')['totalamount'].sum().reset_index()
revenue['rank'] = revenue['totalamount'].rank(ascending = False)
revenue.sort_values('rank')
```

Out [78]:

	roomtype	totalamount	rank
4	suite	1308000.0	1.0
0	cottage	693000.0	2.0
1	deluxe	560000.0	3.0
3	standard	534000.0	4.0
2	dormitory	127800.0	5.0

In [79]:

```
import sqlite3
conn = sqlite3.connect('resort.db')
bookings.to_sql('booking_table',conn, if_exists = 'replace', index = False)
```

Out[79]: 352

In [80]:

```
pd.read_sql_query("select * from booking_table where roomtype = 'suite' and rating >= 4 ",conn)
```

Out[80]:

	bookingid	checkindate	guestid	roomtype	nights	guestscount	totalamount	source	rating	base_price
0	9000	2025-10-26 00:00:00	2025	suite	2.0	suite	12000.0	direct	5.0	6000
1	9001	2025-01-01 00:00:00	2035	suite	1.0	suite	6000.0	booking.com	5.0	6000
2	9003	2025-01-15 00:00:00	2046	suite	4.0	suite	24000.0	direct	4.0	6000
3	9004	2025-09-13 00:00:00	2006	suite	4.0	suite	24000.0	makemytrip	4.0	6000
4	9012	2025-04-05 00:00:00	2029	suite	4.0	suite	24000.0	direct	4.0	6000
5	9014	2025-12-07 00:00:00	2054	suite	4.0	suite	24000.0	direct	4.0	6000
6	9015	2025-02-25 00:00:00	2006	suite	1.0	suite	90000.0	booking.com	4.0	6000
7	9018	2025-10-05 00:00:00	2013	suite	4.0	suite	24000.0	makemytrip	4.0	6000
8	9021	2025-02-02 00:00:00	2022	suite	2.0	suite	12000.0	booking.com	5.0	6000
9	9024	2025-08-27 00:00:00	2051	suite	4.0	suite	24000.0	direct	4.0	6000
10	9026	2025-03-19 00:00:00	2016	suite	2.0	suite	12000.0	direct	5.0	6000
11	9033	2025-10-18 00:00:00	2024	suite	3.0	suite	18000.0	makemytrip	4.0	6000
12	9053	2025-09-16 00:00:00	2010	suite	2.0	suite	12000.0	makemytrip	5.0	6000
13	9063	2025-08-06 00:00:00	2043	suite	4.0	suite	24000.0	goibibo	4.0	6000
14	9071	2025-03-26 00:00:00	2005	suite	2.0	suite	12000.0	direct	5.0	6000
15	9074	2025-11-12 00:00:00	2016	suite	2.0	suite	12000.0	direct	4.0	6000
16	9075	2025-04-04 00:00:00	2011	suite	4.0	suite	24000.0	direct	4.0	6000
17	9082	2025-01-18 00:00:00	2010	suite	2.0	suite	12000.0	makemytrip	4.0	6000
18	9085	2025-06-05 00:00:00	2034	suite	1.0	suite	6000.0	makemytrip	4.0	6000
19	9101	2025-05-01 00:00:00	2011	suite	2.0	suite	12000.0	direct	5.0	6000
20	9112	2025-11-27 00:00:00	2019	suite	2.0	suite	12000.0	direct	5.0	6000
21	9124	2025-01-13 00:00:00	2014	suite	3.0	suite	18000.0	direct	5.0	6000

22	9128	2025-03-28 00:00:00	2029	suite	4.0	suite	24000.0	makemytrip	4.0	6000
23	9133	2025-03-17 00:00:00	2047	suite	3.0	suite	18000.0	booking.com	4.0	6000
24	9146	2025-11-24 00:00:00	2034	suite	3.0	suite	18000.0	makemytrip	4.0	6000
25	9153	2025-06-30 00:00:00	2030	suite	4.0	suite	24000.0	makemytrip	4.0	6000
26	9156	2025-05-24 00:00:00	2016	suite	3.0	suite	18000.0	direct	4.0	6000
27	9168	2025-01-04 00:00:00	2054	suite	2.0	suite	12000.0	makemytrip	5.0	6000
28	9170	2025-12-13 00:00:00	2012	suite	1.0	suite	6000.0	direct	5.0	6000
29	9174	2025-04-18 00:00:00	2028	suite	3.0	suite	18000.0	booking.com	5.0	6000
30	9175	2025-01-27 00:00:00	2019	suite	3.0	suite	18000.0	direct	5.0	6000
31	9185	2025-03-13 00:00:00	2036	suite	3.0	suite	18000.0	makemytrip	4.0	6000
32	9202	2025-01-08 00:00:00	2039	suite	2.0	suite	12000.0	direct	4.0	6000
33	9210	2025-02-14 00:00:00	2052	suite	2.0	suite	12000.0	makemytrip	4.0	6000
34	9211	2025-10-24 00:00:00	2058	suite	2.0	suite	12000.0	booking.com	4.0	6000
35	9218	2025-04-20 00:00:00	2060	suite	3.0	suite	18000.0	direct	4.0	6000
36	9219	2025-10-31 00:00:00	2009	suite	2.0	suite	12000.0	booking.com	4.0	6000
37	9227	2025-06-12 00:00:00	2013	suite	1.0	suite	6000.0	goibibo	4.0	6000
38	9228	2025-12-01 00:00:00	2031	suite	1.0	suite	6000.0	booking.com	4.0	6000
39	9244	2025-09-27 00:00:00	2012	suite	3.0	suite	18000.0	direct	4.0	6000
40	9261	2025-05-03 00:00:00	2007	suite	4.0	suite	24000.0	makemytrip	4.0	6000
41	9269	2025-09-22 00:00:00	2044	suite	2.0	suite	12000.0	direct	4.0	6000
42	9275	2025-08-11 00:00:00	2003	suite	2.0	suite	12000.0	booking.com	5.0	6000
43	9278	2025-09-19 00:00:00	2057	suite	2.0	suite	12000.0	goibibo	5.0	6000
44	9281	2025-04-14 00:00:00	2046	suite	2.0	suite	12000.0	goibibo	5.0	6000
45	9282	2025-11-10 00:00:00	2031	suite	4.0	suite	24000.0	makemytrip	5.0	6000
46	9294	2025-08-20 00:00:00	2019	suite	4.0	suite	24000.0	booking.com	4.0	6000
47	9296	2025-09-13 00:00:00	2029	suite	4.0	suite	24000.0	goibibo	5.0	6000
48	9300	2025-06-03 00:00:00	2050	suite	2.0	suite	12000.0	goibibo	4.0	6000
49	9325	2025-10-27 00:00:00	2027	suite	2.0	suite	12000.0	makemytrip	4.0	6000
50	9327	2025-04-06 00:00:00	2057	suite	1.0	suite	6000.0	goibibo	5.0	6000
51	9329	2025-02-03 00:00:00	2028	suite	1.0	suite	6000.0	makemytrip	4.0	6000

52	9336	2025-11-02 00:00:00	2038	suite	4.0	suite	24000.0	makemytrip	4.0	6000
53	9341	2025-02-25 00:00:00	2036	suite	1.0	suite	6000.0	direct	5.0	6000
54	9342	2025-04-09 00:00:00	2046	suite	1.0	suite	6000.0	makemytrip	5.0	6000
55	9343	2025-03-06 00:00:00	2019	suite	4.0	suite	24000.0	direct	4.0	6000
56	9344	2025-11-18 00:00:00	2060	suite	2.0	suite	12000.0	direct	5.0	6000
57	9347	2025-04-04 00:00:00	2008	suite	3.0	suite	18000.0	makemytrip	4.0	6000

In [59]:

```
# sql show guest name where there spending is more than 10000
guests.to_sql('guests_table', conn, if_exists = 'replace', index = False)
pd.read_sql_query("""select g.guestname, sum(totalamount) as total_spent from booking_table b join guests_table
```

Out[59]:

	guestname	total_spent
0	ravi verma	243200.0
1	karan chauhan	159400.0
2	karan verma	122000.0
3	divya sharma	120800.0
4	pooja singh	115000.0
5	rohan mishra	98200.0
6	rohan gupta	93000.0
7	simran verma	86800.0
8	tanya sharma	86500.0
9	ritika chauhan	86000.0
10	sameer yadav	85700.0
11	simran singh	84000.0
12	yash singh	79500.0
13	tanya verma	75600.0
14	isha mishra	71400.0
15	ravi kumar	68900.0
16	payal chauhan	65500.0
17	yash mishra	64600.0
18	neelam sharma	60600.0
19	tanya kumar	58500.0
20	pooja chauhan	57000.0
21	rohan chauhan	56600.0
22	karan mishra	56100.0
23	divya gupta	55500.0
24	simran sharma	55000.0
25	sameer mishra	54900.0
26	simran chauhan	54500.0
27	aarti mishra	54000.0
28	ravi singh	53600.0
29	neelam verma	52200.0
30	komal gupta	51600.0
31	simran yadav	51100.0
32	nikhil gupta	50200.0
33	ankit singh	50200.0

34	ritika mishra	46700.0
35	karan singh	39800.0
36	payal sharma	39600.0
37	rohan sharma	39500.0
38	ankit chauhan	39500.0
39	ankit sharma	36700.0
40	harsh yadav	34000.0
41	neelam chauhan	33200.0
42	gaurav mishra	31500.0
43	isha yadav	31100.0
44	ankit gupta	28900.0
45	sameer kumar	27300.0
46	ritika verma	24900.0
47	divya verma	22200.0
48	aarti singh	18500.0
49	payal yadav	18000.0
50	komal kumar	12700.0
51	isha verma	11000.0

```
In [60]: room_summary = bookings.groupby('roomtype')['totalamount'].sum().reset_index()
source_summary = bookings.groupby('source')['totalamount'].sum().reset_index()
weekday_summary = bookings.groupby('weekday')['totalamount'].sum().reset_index()

with pd.ExcelWriter("noobtopro3rd.xlsx") as writer:
    full_data2.to_excel(writer,sheet_name='clean data',index=False)
    room_summary.to_excel(writer, sheet_name='roomtype summary', index = False)
    source_summary.to_excel(writer,sheet_name = 'source summary',index = False)
    weekday_summary.to_excel(writer,sheet_name = 'weekday summary',index = False)
print("saved")
```

saved

In []: